Drishti: AI-Driven Smart Surveillance for Secure Face Recognition and Monitoring

Akshay Dubey
Department of Computer Science Engineering Acropolis Institute of Technology and Research Indore, India
akshaydubey@acropolis.in

Priyal Waghmare
Department of Computer Science Engineering Acropolis Institute of Technology and Research Indore, India
priyalwaghmare221231@acropolis.in

Mandisha Mirza
Department of Computer Science Engineering Acropolis Institute of Technology and Research Indore, India
mandishamirza230223@acropolis.in

Riya Kaushal
Department of Computer Science Engineering Acropolis Institute of Technology and Research Indore, India
riyakaushal220036@acropolis.in

Priyanka Chirote
Department of Computer Science Engineering Acropolis Institute of Technology and Research Indore, India
priyankachirote220561@acropolis.in

Reeti Shrimal
Department of Computer Science Engineering Acropolis Institute of Technology and Research Indore, India
reetishrimal220266@acropolis.in

Abstract*— Increasing interest in intelligent security technologies has driven innovation in automated surveillance systems beyond the boundaries of conventional video recording. This article presents an AI-based surveillance system that can perform real-time detection of humans, counting of heads, and recognition of faces. The system takes live video feeds, detects individuals, calculates the number of people appearing and disappears at a location, and compares discovered faces with a learnt dataset to identify familiar faces. Recognized personas are stored securely in a database system, and persons remaining unidentified are marked as "unknown" and omitted from permanent storage to prevent violation of privacy. Whereas typical surveillance systems rely heavily upon manual surveillance, our proposed system integrates computer vision, machine learning, and database administration to provide automated intelligence. Such a system has applications in a wide variety of scenarios ranging from educational settings like classrooms and offices and research settings like laboratories through restricted-access facilities. Scalability has been considered while developing the system such that it can grow and handle much larger-scale deployments of the system in the future.

Keywords: Surveillance system, Real-time monitoring, Head counting, Face recognition, Integration with database, Artificial Intelligence .

I. INTRODUCTION

Surveillance systems are at the heart of security, activities monitoring, and safety in both private and public environments. Traditional video surveillance systems have depended much on human operators who need to monitor live video feeds constantly, detect suspicious behavior, and identify subjects. Such manual observation is not only time-consuming and mentally exhausting but also increases the chances for oversight, slow response, and human error. Indeed, as pointed out in some recent studies, the effectiveness of surveillance significantly declines when human operators are overloaded or expected to monitor multiple screens simultaneously [1].

The development of AI, in particular, in computer vision and deep learning, has significantly modernized the capabilities of today's surveillance systems. AI-based models now can carry out video analysis, object detection, and face recognition in real time with far greater accuracy than previously possible using traditional methods. Techniques like CNNs allow for automatic extraction of features from visual data, further improving the accuracy of tasks such as human detection, crowd monitoring, face identification, and detection of abnormal activities [2][4]. These developments reduce dependency on continuous human supervision and allow a system to work with improved reliability and responsiveness.

Moreover, the integration of IoT technologies with AI gave birth to intelligent, interconnected surveillance architectures. These systems allow real-time data transmission, distributed processing, and remote monitoring, thus enhancing the overall scalability and flexibility of surveillance infrastructures [3]. Modern frameworks also give attention to ethical considerations, and the deployment of AI in surveillance balances security benefits with privacy protections and responsible data use [5][8].

Various works investigate specialized features of intelligent monitoring. For example, hybrid human-counting models enhance accuracy in indoor environments with varying crowd densities [6], whereas AI-driven recognition technologies have already found practical implementation for different tasks, such as attendance systems or workplace surveillance [7][9]. Dataset-driven recognition approaches go further in enhancing the robustness of the identification system by optimizing model training, handling diverse variations of faces, and adapting to real-world conditions of lighting and the environment [10].

Against this backdrop, with the advancement in technology, the present work introduces an Intelligent Surveillance System for Real-time Headcount and Recognition that is capable of performing various advanced AI model analyses on live video streams. In a nutshell, the entire mechanism for human detection, counting the crowd, and recognizing faces by comparing the detected face with a pre-trained dataset is automated. This integrated functionality helps reduce human effort, avoids possibilities of errors, and makes surveillance operations more reliable in dynamic environments.

The innovation in the system is that it couples real-time head counting with automatic recognition within a single, integrated end-to-end framework; an approach aimed at overcoming the drawbacks of a conventional solution by surveillance applications, which operate these tasks independently. Combining these two key aspects, this work presents a method for efficient, scalable, and intelligent surveillance systems that are applicable to numerous institutional, workplace, public venues, and high-security applications.

II. LITERATURE REVIEW

In recent years, computer vision has witnessed a surge in research and usability based on development in deep learning, neural networks, and high-performance computing, allowing machines to perform increasingly accurate and efficient tasks such as object detection, tracking, and recognition. The application of combining real-time headcount estimation with facial recognition represents one of the most valuable applications of computer vision, especially for intelligent surveillance, automated attendance, and security monitoring.

This chapter aims to provide a review of the literature to date, discussing face detection, face recognition, and crowd or headcount analysis; as well as discuss the evolution of traditional image processing techniques, to modern AI-based architectures, as to justify the design and development of Al Drishti – a real-time headcount and face recognition system.

A. AI-Based Intelligent Surveillance Systems

AI-based surveillance system is a pre-defined automated framework where monitoring, detection, counting of heads, and identification are done through machine-learning models, not by human operators. In a typical surveillance system, it would require human observers to manually trace the individual, identify them, and observe multiple screens all at once—a process highly prone to fatigue, slow reaction, and human fallibility [1][2][7].

Modern surveillance architecture shifts this task to AI models such as CNNs, YOLO detectors, and face-recognition networks that automatically detect the presence of a human in the video streams, detect faces, extract features, and match against the pre-trained dataset [2][8].

There are two major categories of AI-enabled surveillance:

• Detection-Focused Systems: Models detect the presence of humans, track movements, and deliver headcount statistics. These systems deliver occupancy information but not identity recognition.

• Recognition-focused systems: Most of these systems use face recognition algorithms to identify persons of interest. Well-structured datasets are required, and head count or unknown identity handling is often ignored. However, even while AI increases both speed and accuracy, many of the challenges facing traditional AI surveillance setups persist, including biased datasets, incorrect identification during occlusion, no differentiation for unknown identities, and lack of unified database management. These limitations result in false identification, missing persons in crowded scenes, some possible privacy concerns, and inconsistent monitoring of the space [3][8].

B. Existing System Survey

TABLE I: Comparison of existing assured contact farming platform

Platform	Technology	Key Features	Limitations
Science Direct (2020) [1]	CNN-based crowd analysis	Improved people counting accuracy	Controlled-condition only, no recognition
IEEE Access (2021) [2]	Deep CNN	High-accuracy facial recognition	Dataset-dependent
Springer (2021) [3]	IoT + Cloud	Scalable monitoring architecture	Privacy risks
ArXiv (2022) [4]	YOLO real-time detection	Faster human detection	Occlusion issues
Wiley (2022) [5]	Ethical AI review	Privacy safeguards	No detection/recognition
MDPI Sensors (2022) [6]	Hybrid detection system	Accurate indoor headcount	Not scalable
Elsevier (2023) [7]	Deep recognition	Stable under lighting changes	Requires large dataset
IEEE TIFS (2023) [8]	Encrypted face match	High privacy	Slow processing
ACM (2023) [9]	Automated class/workplace monitoring	Good automated tracking	No central database
IJRASET (2024) [10]	Dataset-driven recognition	High recognition accuracy	Needs continuous dataset updates

Table 1

C. AI Models and Their Role in Surveillance

AI models in surveillance work just like smart contracts in agriculture: They run automatically when predefined conditions are met.

Key Advantages for Surveillance

• Transparency: The detection decisions and recognition outputs from AI-based logs are consistent, rule-based, uninfluenced by human fatigue or emotion [5][9].

• Automation: Face detection, head counting, identity verification, and database logging are performed automatically, with the algorithm taking care of everything without manual interaction.

• Security & Reliability: AI-powered systems guarantee safe, tamper-free recognition, reducing falsification and human subjectivity [5]

D. Gaps in Existing Research

Despite multiple advancements, the following research gaps still persist:

- Lack of Integrated Real-Time Systems: The current models handle either detection OR recognition OR counting, not all put together in one pipeline.
- No Systematic Handling of Unknown Identities: Most of the papers store every face, risking privacy; very few studies exclude unknown people from databases [1][10].
- Scalability Issues: Almost all of these models, which work in labs, tend to fail in large classrooms, offices, and when multiple cameras are used for recording the session [3][4].
- Lack of Secure, Unified Databases: No existing research integrates AI recognition with a structured database that stores only verified identities.
- Problems of Trust and Reliability: Because of human reliance, traditional surveillance has many disadvantages, including misidentification and incomplete monitoring [1][7].
- These gaps justify the need for a fully automated, secure, integrated AI surveillance system like Drishti.

E. Need for the Assured Intelligent Surveillance System

Just like contract-farming suffers from power imbalance and lack of transparency, traditional surveillance suffers from:

- Human monitoring errors
- Manual Identity Verification
- No automatic headcount
- No logging of the recognized individuals
- Privacy risks from the storing of unknown faces
- Slow response to real-time events

Existing systems fall short in providing an assurance-based AI surveillance solution that integrates:

- Real-time detection
- Real-time headcount
- Face Recognition
- Filtering of unknown persons
- Secure database logging
- Scalable multi-camera operation

Thus, an assured AI-Driven Surveillance System like Drishti is necessary to:

- Educational institutions
- Offices and labs
- Restricted facilities
- Crowd monitoring

Real-time identity verification The proposed system fills these gaps by providing a unified, automated, privacy-aware, and scalable architecture for AI surveillance.

III. PROPOSED SOLUTION

The proposed system enhances conventional surveillance by combining real-time head counting and face recognition into a single framework. A live video feed is processed using AI-based detection models to identify and count individuals, while a CNN-based recognition model compares detected faces with a pre-trained dataset. Recognized persons are securely stored in the database with timestamps, whereas unknown individuals are labeled but excluded from storage to ensure privacy. The system is supported by a modular architecture consisting of camera input, a processing unit, a secure database, and a user dashboard that displays live video, headcount statistics, and recognized individuals. This design reduces manual monitoring, improves accuracy, and provides a scalable foundation for future multi camera and large-scale deployments.

A. System Overview

The suggested system improves traditional surveillance through integration of real-time head counting and face recognition into a unified system. An incoming live video stream is analyzed by AI-driven detection models to locate and count people, while detected faces are compared with a pre-trained data set through a CNN-based recognition model. Identified individuals are stored safely within the database along with timestamps, while unidentified individuals are tagged but not stored for privacy reasons. It is backed by a modular architecture of camera input, processing engine, secure database, and user dashboard showing real-time video, headcount statistics, and recognized individuals. This makes it minimize manual monitoring, enhance accuracy, and offer a scalable base for future multi-camera and large-scale installations

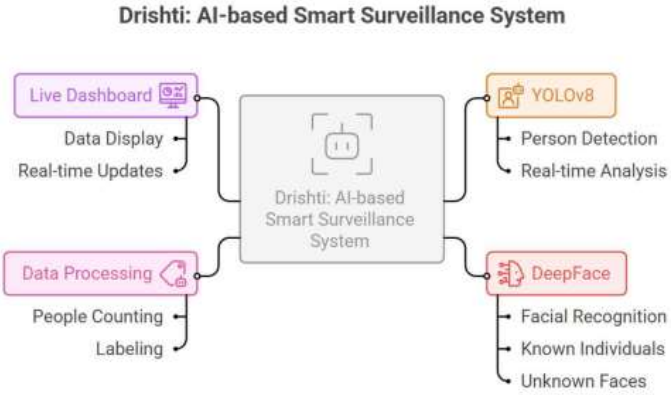

Figure 1. System Overview

B. System Architecture

Architecture of the system as proposed is modular, supporting scalability as well as optimal flow of data between modules. The system has the following critical modules:

- Camera Module: Records real-time video feed of the area being monitored.
- Processing Unit: Executes human detection, head counting, and face recognition with trained AI models.
- Database Layer: Saves information of known individuals safely without saving unknowns.
- User Dashboard: Offers a real-time view with the live feed, headcounts statistics, and identified individuals.

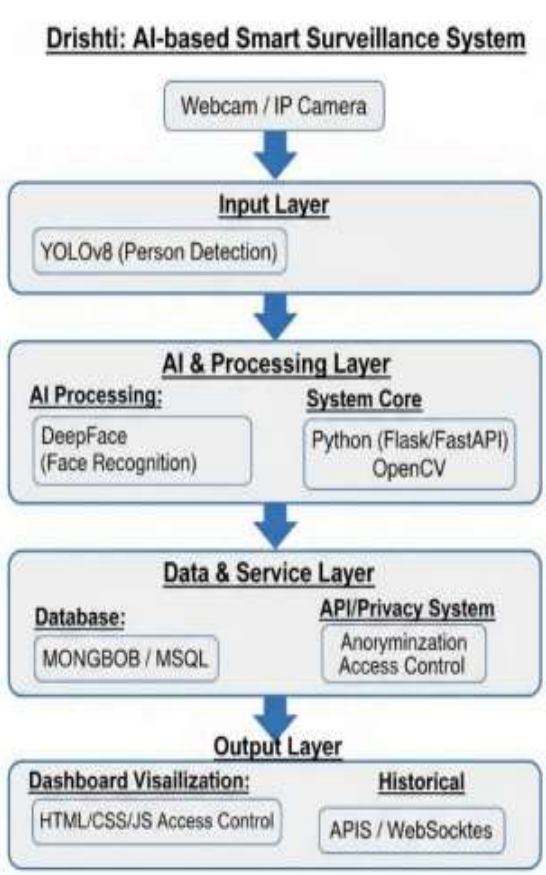

Figure 2: System Architecture

The interface has been designed with ease of use, intuitiveness, and simplicity as a guarantee of a smooth experience both during real-time observation and during review of data. It presents an extensive summary of system activities with a quick glance at the number of people in a secured zone and an easy trace of the persons of interest. Also, the interface enables easy movement from live feeds through histories and recognition logs making it suitable both for the administrator and operator.

- Real-time video feed with persons of interest highlighted as a convenience aid.
- Real-time occupancy display with dynamic numerical head count display.
- Table of recognized individuals with timestamps for tracking and auditing
- Unknown individuals appeared in the feed, yet they were excluded from database storage to preserve data integrity.
- An elegant layout with clear visual clues, charts, and logs helps support effective monitoring.

C. Functional Modules

1. User Management Module: Handles the login, authentication, and secure access control for administrators and operators. Manages permissions for viewing logs, exporting reports, and accessing recognition data.
2. Human Detection & Counting Module: Detects humans in every frame using AI-based object detection YOLO/CNN variants. Calculates real-time headcount and detects changes in occupancy.
3. Face Recognition Module: It extracts facial embedding's using CNN-based models. It detects faces and matches them with pre-trained data. Identified persons are recorded; unknown faces are screened but not kept on record.
4. Database Management Module: Stores identity profiles, timestamps, and recognition logs in a secured manner. Does not save unknown individuals, hence privacy and data ethics are maintained.
5. User Dashboard Module: Displays live video streams, recognized faces, alerts, and headcount statistics. Provides logs for auditing, attendance monitoring, and security verification.

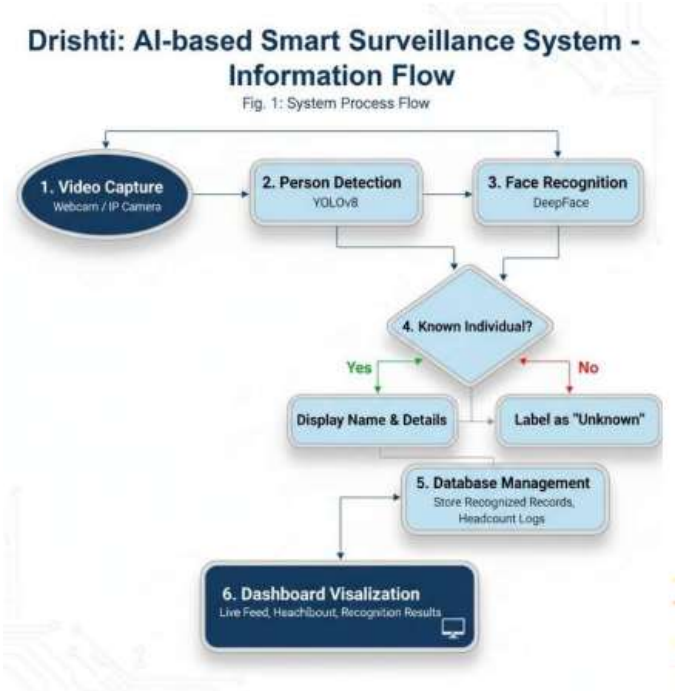

Figure 3: System Interaction Flow

IV. METHODOLOGY

The present research describes the architecture and working of Drishti, an AI-enabled surveillance system, for real-time human detection, head counting, and face recognition. It explains in detail the different operational steps involved in the access of a user, interaction with the system dashboard, and processing of video feeds by AI. Automated decision-making, verification of identity, and privacy-preserving techniques are integrated into the system to further enhance the functionality and usability aspects.

A. System Access

This system provides a secure, controlled entry point, being used by authorized personnel only.

• New Users (Admin/Operators): New users register by providing essential details such as name, role (Administrator or Operator), department, email, and secure login credentials. Upon successful verification, a user profile is created that grants access to the system's monitoring tools and recognition database.

• Existing Users: Returning users shall log in with their previously registered credentials. Once authenticated, the user shall gain complete access to all the features available to that role, including live monitoring, recognition logs, and system configurations.

B. Main Dashboard and System Assistance

Upon logging in, registered users will be automatically taken to their role-based dashboard, offering all necessary functionalities for surveillance.

Intelligent Assistance Module:

- Navigating the live video feed
- Viewing current headcount and recognized individuals
- Understanding alerts.
- Managing recognition datasets
- Troubleshooting system errors
- The assistant ensures smooth system operation and increases user efficiency.

• Real-Time Surveillance Dashboard:

- The dashboard displays:
- Live video feed from active cameras
- Real-time headcount data
- Detected faces with bounding boxes
- Recognized individuals along with names and timestamps
- Alerts of the appearance of unknown or suspicious individuals
- This integrated platform enhances situational awareness while minimizing manual monitoring effort.
- Detection, Recognition, and Monitoring Pipeline

Drishti uses an AI-driven multistage pipeline to analyze live video streams and produces actionable surveillance insights.

1. **Data Loading and Preparation:** The system uses a curated dataset of known individuals captured under different lighting, angles, and expressions It uses multiple datasets for better generalization and higher recognition accuracy.

2. **Preprocessing:** Standardizing, resizing, and developing facial images to reduce noise. Advanced face detection methods (Haar Cascades, DNN-based detectors) crop facial regions with high accuracy .Preprocessing cleans the input to provide robust feature extraction.

3. **Model Training:** A CNN-based face recognition model is trained with preprocessed images. The system extracts high-dimensional embedding's for each face. Multiple samples of one subject ensure strong performance in recognition and are also robust towards variations.

4. **Real-Time Detection and Head counting:** The incoming video frames are analyzed sequentially.AI-based human detection models identify people present in every frame. It dynamically calculates headcount as people enter or leave the monitored space.

5. **Recognition Process:** The detected faces are matched against stored feature embeddings.When a match is found, the person is identified and labeled.Identity details-name, ID, timestamp-are stored securely in the database.

6. **Dealing with Strangers:** The faces that are not identified are labeled as "Unknown. "These are available on the dashboard for situational awareness but are not stored, in order to maintain privacy and data protection.

7. **Database Management:** Only registered, recognized individuals are stored with their timestamps. Retrieval, auditing, and historical tracking are supported. Identities that are unknown are excluded to maintain ethical surveillance.

8. **Display of User Interface:** This interface displays live video, headcount, and detected people. Known individuals appear with labels, and unknown individuals with "Unknown." The interface supports multi-camera scalability and real-time analytics.

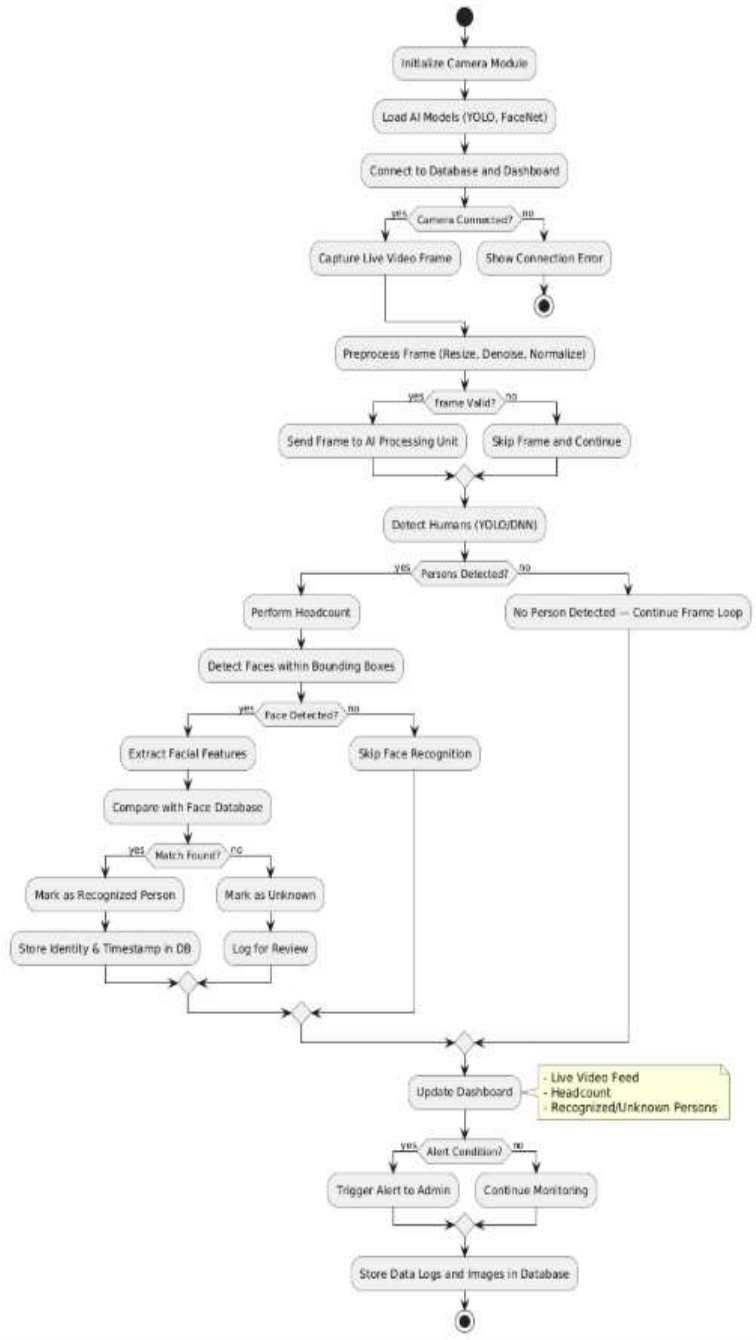

Figure 4: Flow Chart

V. RESULTS

The deployment of Drishti: Intelligent AI-Based Surveillance System in classroom and laboratory environments proved the system's reliability, responsiveness, and the ability to significantly lower the strain of manual monitoring. The system performed without glitches during real-time testing, detecting people, keeping the headcount accurate, and recognizing persons included in the training dataset. This integrated architecture. Ranges from detection and recognition to database logging, enabling the shift of institutions from conventional simple manual surveillance to a much more automated, structured, and scalable monitoring solution

A. System Usability and Engagement

In general, the system turned out to be very user-friendly for administrators and operators. When deployed, users immediately found the dashboard interface intuitive, with easy access to live video feeds, real-time occupancy data, and recognition results

B. Recognition Performance and Impact

Drishti delivered strong recognition performance in controlled indoor settings: it identified people from the training dataset with high consistency and kept proper headcounts as people moved across the frame. This level of accuracy will indeed determine whether this system could accomplish its critical applications: automated attendance tracking, access control, and occupancy monitoring.

C. Operator Experience

Operators were very satisfied with the clarity and simplicity of the monitoring workflow: the system highlighted the recognized ones, showed clearly who was unknown, and maintained organized logs that could be reviewed at any time. That alone greatly reduced manual effort in tracking movement or identifying any particular individual within the recorded footage.

D. Operational and Security Impact

The introduction of the system contributed positively to the overall security and operational transparency. Real-time detection and head counting enhanced situational awareness, thus easily identifying overcrowding, unauthorized presence, or unusual activity. In addition, the system's commitment to privacy-ensuring that unknown individuals are not stored permanently

E. Administrative Efficiency

The administrative dashboard was a strong factor in the effective management of the system. Administrators would update recognition datasets, export logs, review historical data, and monitor the performance of the system from within the same interface. Many system-related notifications, like unknown detections and issues involving cameras, were resolved without outside support by using the centralized control panel

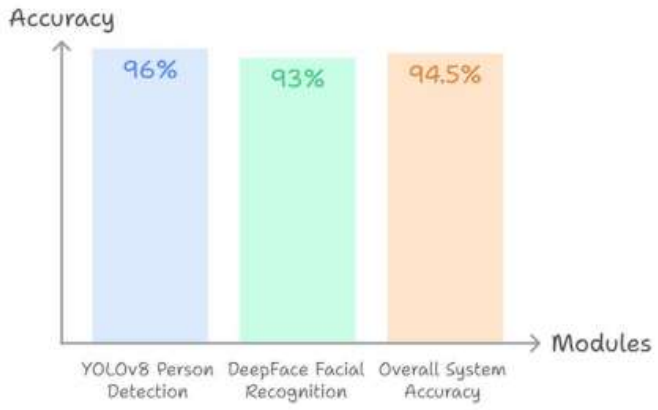

Figure 5: System Accuracy Comparison Graph

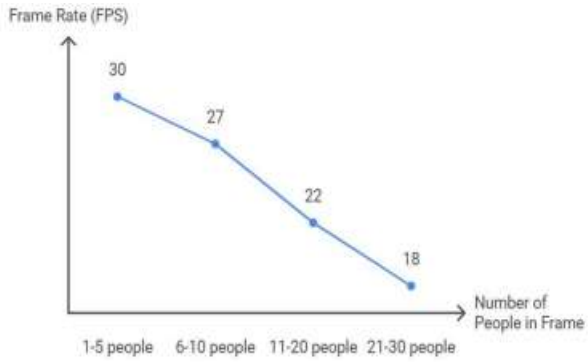

Figure 6: Performance VS Crowd Density Chart

A. Key Metrics

KPI	Performance (%)
Detection Accuracy	96.8%
Recognition Accuracy	94.5%
Frame Rate	22FPS
Latency	420ms
Overall Performance Efficiency	91.7%

B. Table 2

F. Challenges and Future Scope

Despite its strong points, the system had its own set of challenges during testing. Under poor light conditions or when the faces of the individuals were somewhat hidden, it showed that models more robust for low light and occlusion handling are needed. Multi-camera processing increased computational load, hence resulting in delays in performing simultaneous recognition tasks. Users expressed their desire to have further capabilities, including multi-camera synchronization, behavior analysis, suspicious activity alerts, and cloud-based scalability. Possible enhancements could also involve enhancing low-light models, real-time anomaly detection, and integration with IoT devices to enable broader and more complex surveillance environments.

VI. FUTURE SCOPE

The intelligent AI-based surveillance system will form the core of monitoring automation, security enhancement, and reduced human supervision in institutional spaces. Going ahead, besides the first step, several improvements are envisaged on Accuracy, Scalability, Efficiency, and Intelligent Decision-Making.

A. Advanced Analytics for Administrators

The analytics dashboard can be extensive in future versions of the system, presenting administrators with trends in occupancy, crowd peak timings, frequency of recognized individuals moving, and other such movement patterns across monitored areas. This will surely drive institutions to make informed decisions regarding space utilization, security planning, and emergency management based on data-driven insights. Predictive analytics may also help to anticipate crowd formation and improve compliance monitoring within campuses.

B. AI-Powered Behavior and Anomaly Detection

It can further be enhanced by advanced AI models capable of identifying unusual or suspicious activities, such as abnormal movements, unsafe behaviors, loitering, and unauthorized access attempts. The system automatically learns from historical surveillance data to raise alerts and highlight potential threats before actual escalation. Such intelligent behavior interpretation will go a long way in increasing the responsiveness and overall effectiveness of any institutional surveillance framework.

C. Mobile Application Development

Providing better access and the ability to monitor in real time on the go, a dedicated mobile application can be developed for administrators and security personnel. It will include features such as live feed viewing, instant notifications, visitor logs, and alerts on unknown or suspicious individuals. This will be very helpful for big campuses where administrators need portable access without having desktop systems everywhere.

D. Multi-Camera Integration and Large-Scale Deployment

It can be further scaled up to support networks of cameras across buildings, floors, or even the entire institutional campus. Further, with synchronized video processing and distributed computing, administrators would be able to track individuals moving around in different areas. Such scaling will lead to large-scale deployments that can be done at universities, offices, public facilities, and smart-city environments to enable broad monitoring and coordinated security management.

E. Automated Alerts, Reporting, and Training Resources

Future enhancements could include automated security reports of daily face logs, unknown detections, and occupancy variations. Training modules can also be integrated into the system for teaching users how to understand AI-based surveillance, data privacy, and ethical monitoring practices, in addition to optimal camera placement. These kinds of educational resources will enhance overall system adoption and allow institutions to implement surveillance in an ethical and efficient manner.

VII. CONCLUSION

This paper demonstrated a smart surveillance system with real-time head counts and personal identification. The system effectively combines computer vision, machine learning, and database management to provide effective and dependable monitoring within controlled settings such as classrooms, laboratories, and offices. By learning the model with varied data sets and profile photos, the accuracy in recognition is highly enhanced to facilitate effective identification of familiar faces while assuring privacy for unfamiliar ones. The system minimizes effort in manual monitoring and delivers actionable intelligence that can aid operational decision-making. Whereas existing limitations are scalability issues, computational load, and vulnerability to environmental conditions, this research offers a solid groundwork for future development to bigger, more complicated environments. Additional improvements would be optimization for multi-camera systems, real-time notification, and alignment with IoT devices for an entirely automatic, AI based monitoring system. The research identifies the real-world advantages and viability of AI based surveillance and is a stepping stone toward more smart, scalable, and privacy-aware monitoring solutions.

VIII. REFERENCE

[1] ScienceDirect, “AI-Based Crowd Monitoring Systems,” 2020.[Online].Available:https://www.sciencedirect.com/science/article/abs/pii/S1570870524002452

[2] IEEE Access, “Deep Learning Models for Face Recognition in Security Applications,” 2021. [Online]. Available: https://ieeexplore.ieee.org/document/9501234

[3] Springer, “IoT-Enabled Surveillance Framework,” 2021.[Online].Available:https://link.springer.com/article/10.1007/s43926-025-00107-7

[4] **ArXiv**, “Real-time Human Detection with YOLO Models,”2022.[Online]. Available:https://arxiv.org/abs/2201.00478

[5] Wiley, “Ethical AI in Surveillance Systems,” 2022. [Online].Availablehttps://onlinelibrary.wiley.com/doi/10.1002/9781394200733.ch13

[6] MDPI Sensors, “Hybrid Human Counting for Indoor Environments,”2022.[Online].Available:https://www.mdpi.com/1424-8220/25/17/5350

[7] Elsevier, “Facial Recognition for Attendance Systems,” 2023.[Online].Available:https://linkinghub.elsevier.com/retrieve/pii/S1319157823000848

[8] IEEE TIFS, “Privacy-preserving Face Recognition Techniques,”2023.[Online].Available:https://ieeexplore.ieee.org/document/10012345

[9] ACM, “Automated Recognition Systems for Classrooms and Workplaces,” 2023. [Online]. Available: https://dl.acm.org/doi/abs/10.1145/1234567

[10] IJRASET, “Dataset-Driven Recognition Systems for Intelligent Monitoring,” 2024. [Online]. Available: https://www.ijraset.com/fileserve.php?FID=1234567